11 January 1974
SUPERSEDING
MIL-C-41836B
27 August 1969

MILITARY SPECIFICATION

CLOTH, DUCK; COTTON AND NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of cotton and nylon duck cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1135 - Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-Synthetic Fiber Blends)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 8305

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PUBLICATIO .

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.5).

3.3 Material.

3.3.1 Cotton. The cotton shall be carded and drawn.

3.3.2 Nylon. The nylon shall be semi-dull staple having a uniform cut length of 1-1/2 inches and a round cross section with a single nominal denier of 2.25. The use of any form of nylon waste is prohibited, e.g., undrawn fiber, mixture of deniers, luster or cross sections, and waste from any stage of fiber production whether drawn, undrawn or mixed or garnetted fiber (see 4.6).

3.3.3 Yarn ply. The nylon and cotton staple shall be blended, spun and twisted into a 3-ply yarn for the warp and filling so that the finished cloth shall contain not less than 45 percent cotton, and not less than 50 percent, or more than 55 percent nylon when tested as specified in 4.6.

3.4 Color. The color shall be as specified (see 6.2), and shall match the standard sample (see 6.3 and 6.4). The use of disperse dyes on the nylon is prohibited. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.6. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.6.

3.4.1 Matching. The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2300° Kelvin.

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3.4.2 Colorfastness. The dyed and finished cloth shall show colorfastness to accelerated weathering equal to or better than the standard sample when tested as specified in 4.6. When a standard sample is not available, the dyed and finished cloth shall show "good" fastness to accelerated weathering when tested as specified in 4.6.

3.5 Physical requirements. The finished cloth shall conform to the requirements as specified in table I when tested as specified in 4.6.

TABLE I. Physical requirements

Weight, oz. per square yard (minimum)	Yarns per inch (minimum)		Breaking strength, pounds (minimum)		Air permeability, cu. ft./min.sq.ft. (maximum)	Weave
	Warp	Filling	Warp	Filling		
14.5	46	26	500	300	3.0	Plain

3.6 Width. The minimum width inclusive of selvages shall be as specified (see 6.2).

3.7 Finish. The cloth shall be mildew resistant treated with copper 8-quinolinolate so that the treated cloth contains a maximum of 0.15 percent and a minimum of 0.04 percent copper as metal from copper 8-quinolinolate when tested as specified in 4.6.

3.8 pH. The pH value of the water extract of dyed and finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.6.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in full width rolls as specified in PPP-P-1135.

3.10 Face identification. The face side of the cloth shall be identified by stamping the word "FACE" on that side of the cloth at each end of the roll.

3.11 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a) First article inspection (see 4.3).
b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When required (see 6.2), the preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish, and shall be tested in accordance with methods specified in 4.6.

* 4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.4.

* 4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1 with the following exceptions: glossary numbers 1, 2, 3, 5, 14, 26, 27 and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of warp yarns. Only those knots and slubs which exceed three times the normal yarn thickness shall be scored as defects. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be

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20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 53 points. If one roll exceeds 53 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 53 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

* 4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth
Objectionable odor
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Tackiness (sticky to touch)

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

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4.4.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

* 4.4.4 Examination for face marking and compliance with the Rules and Regulations Under the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample

Face stamping missing from either or both ends
Face stamping on wrong side
Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act

* 4.5 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine whether packaging, packing and marking comply with section 5 requirements.

* 4.6 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2 continuous yards full width of the finished cloth. The sample size shall be as listed in table II. The lot shall be unacceptable if one or more units tested fail to meet any of the specified test characteristics. All test reports shall contain the individual values utilized in expressing the final results.

TABLE II. Sample size

Lot size in yards	Sample size
300 or less	2
301 up to and including 22,000	3
22,001 and up	5

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TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Yarn material identification	3.3.1 and 3.3.2	1/
Nylon staple:		
Luster	3.3.2	2/
Denier	3.3.2	2/
Staple length	3.3.2	2/
Absence of nylon waste	3.3.2	2/
Yarn ply	3.3.3	Visual 3/
Fiber content:		
Cotton	3.3.3	4.7
Nylon	3.3.3	4.7
Presence of labile sulfur	3.4	2020
Colorfastness to accelerated weathering	3.4.2	5671 4/
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Air permeability	3.5	5450
Weave	3.5	Visual 3/
Copper 8-quinolinolate	3.7	5/
Copper content	3.7	2050
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ The supplier shall furnish a certificate of compliance furnished by the fiber producer that each lot of nylon staple used conforms to the requirements.

3/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

4/ The time of exposure shall be 40 hours.

5/ The manufacturer shall certify that only copper 8-quinolinolate was used in the treatment of the cloth.

* 4.7 Test for fiber content. The sample for test shall be a 2 to 5 gram sample of the cloth. The fiber content shall be determined on a dry desized specimen in accordance with Method 2530 of FED-STD-191 except that method of calculation shall be as follows:

$$\text{Percent nylon} = \frac{S - R}{S} \times 100$$

$$\text{Percent cotton} = \frac{R}{S} \times 100$$

5. PREPARATION FOR DELIVERY

* 5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1135.

6. NOTES

6.1 Intended use. The cloth is intended for use in the manufacture of tropical combat boots.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

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(b) First article. When preproduction sample is required (see 3.2, 4.3 and 6.5).
(c) Color of cloth required (see 3.4).
(d) Width of cloth required (see 3.6).
(e) Minimum length of roll, if other than specified (see 3.9).
(f) Selection of applicable levels of put-up packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 A suggested but not mandatory dyestuff formulation for Olive Green 107 is as follows:

Cotton:

Vat Green 8, CI No. 71050
Vat Black 25, CI No. 69525
Vat Green 3, CI No. 69500

Nylon:

Acid Green 70
Acid Brown 45
Acid Yellow 151
Acid Orange 64

The standard shade sample for Olive Green 107 has been dyed with the above formulation.

* 6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample consisting of 2 yards of the finished cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.6 Marginal notation. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-41836C CLOTH, DUCK; COTTON AND NYLON

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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